

Tribhuvan University, Institute of Science and Technology



Comparing CatBoost, TabPFN, and TabFM for Explainable Hair Fall Risk Prediction

Dissertation Proposal

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Project at a glance

Problem	Hair fall risk is hard to judge, and prediction models are hard to explain.
Objectives	Benchmark CatBoost, TabPFN, and TabFM, and explain every model with SHAP.
Research gap	Foundation models were never tested on hair fall data, and CatBoost was never tuned.
Data and method	21,606 records, 3 risk tiers; same split for all models; accuracy, macro-F1, ROC-AUC.
Expected outcome	A fair comparison and explanations that agree with known causes.

Problem statement

- Hair loss risk is hard to judge: hormones, nutrition, stress, and habits act together.
- Earlier studies used classical algorithms, with accuracies from about 50% to 100%.
- Newer foundation models (TabPFN, TabFM) are untested on hair fall data and hard to explain.

Objectives

1. Implement and benchmark CatBoost, TabPFN, and TabFM for multi-tier hair fall risk.
2. Explain every model with SHAP so each predicted risk tier can be traced to specific indicators.

Background: what causes hair loss

Factor	What it does
Hormones and heredity	Inherited sensitivity to androgens; family history matters
Nutrition and iron	Shortage of iron, zinc, and vitamins weakens follicles
Thyroid function	Thyroid hormones regulate the hair cycle
Psychological stress	Pushes follicles into the resting phase early

Literature: what others did

Study	Models	Main result
Khatun et al. (2022)	SVM, KNN, LR, RF, XGBoost	XGBoost 92.62%
Kumar et al. (2025)	RF, XGBoost, CatBoost, LightGBM	RF 100%, CatBoost 49.5%
Siami & Azis (2025)	LR, DT, RF, GB, XGBoost	About 50% at best

LR = Logistic Regression, DT = Decision Tree, RF = Random Forest, GB = Gradient Boosting

Literature: the three models

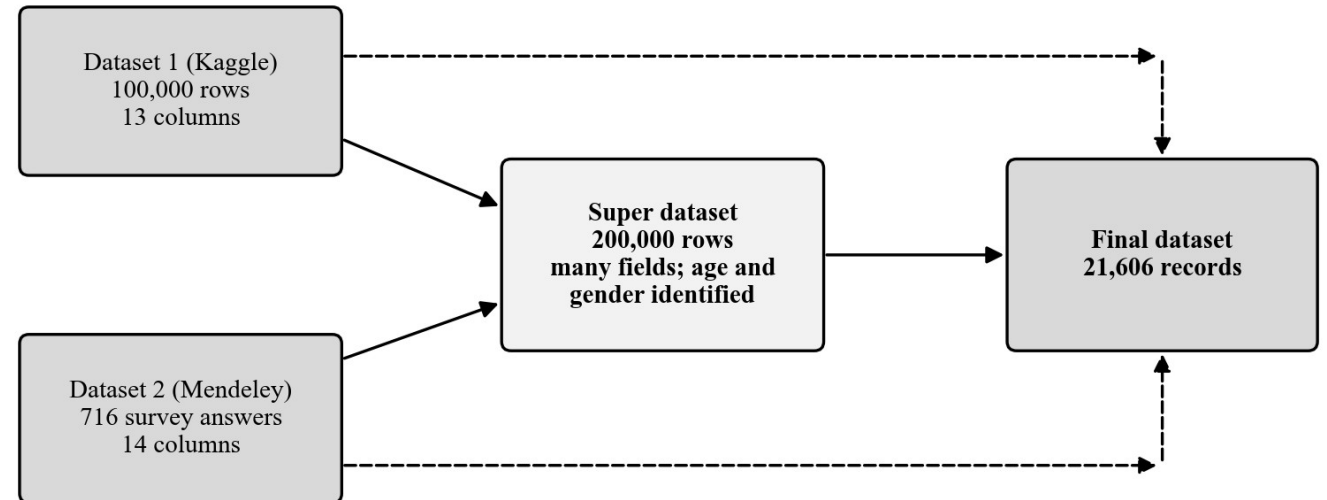
- CatBoost (Prokhorenkova et al., 2018): boosted trees for categorical data.
- TabPFN (Hollmann et al., 2025): pretrained transformer, best on small tables.
- TabFM (Kong et al., 2026): 400-million-parameter foundation model.

Research gap

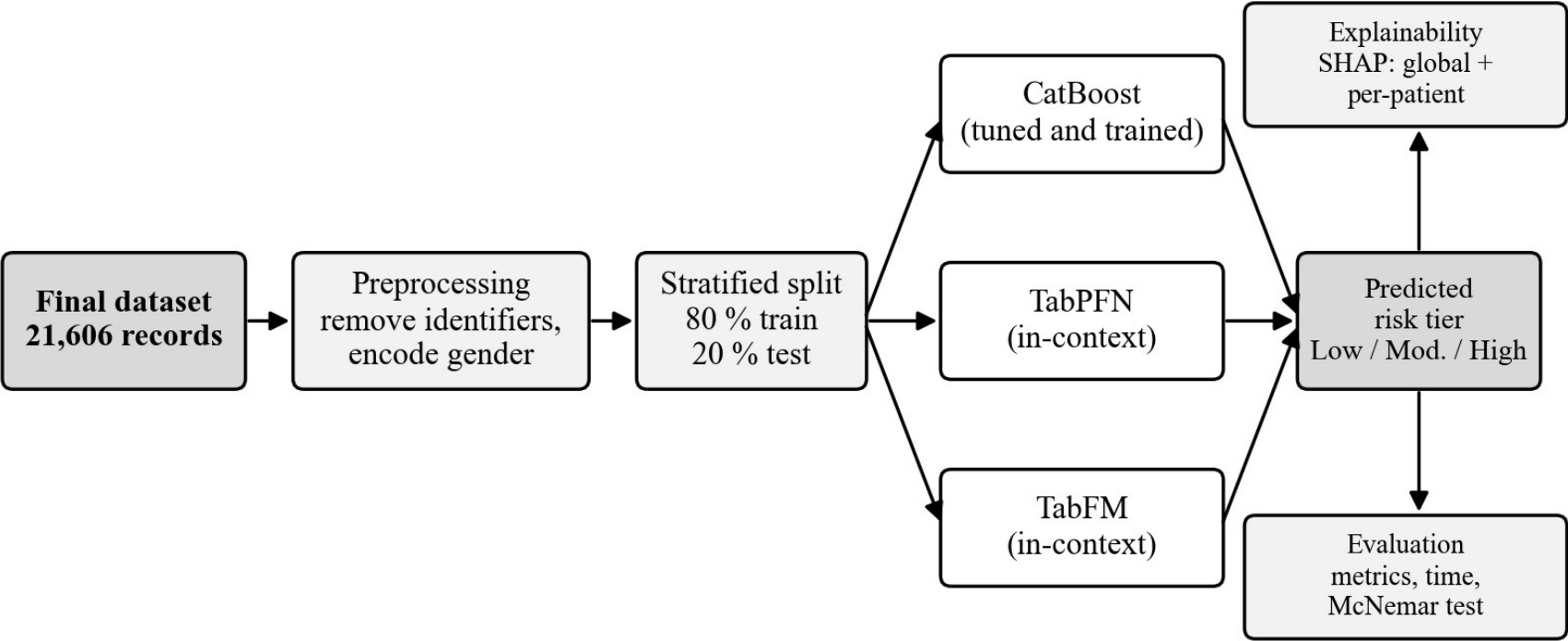
- Only classical algorithms were used, and CatBoost was never tuned.
- TabPFN and TabFM were never tested on hair loss data.
- No study explains boosted trees and foundation models side by side.

Data

- Two public datasets: Kaggle and Mendeley.
- Super dataset: 200,000 records.
- Final dataset: 21,606 records, 20 features, 3 risk tiers.



Research framework



Same split and seed for all models.

Models

Model	Tuning
CatBoost	Trained and tuned (grid search, 5-fold cross-validation)
TabPFN	None
TabFM	None

Expected outcomes

- A fair comparison of the three models on the same data.
- TabPFN and TabFM competitive without tuning (to be tested).
- Explanations that agree with known causes: iron, stress, family history.

Thank you

Questions are welcome

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